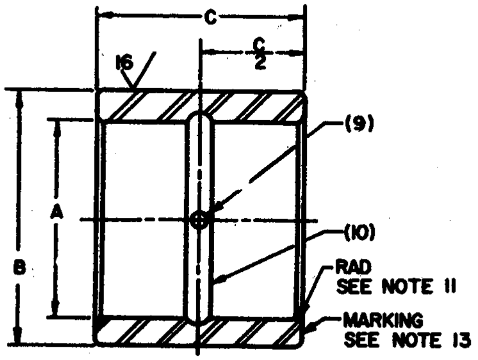
FED. SUP CLASS
3110



MS PART NO.	A BORE DIAMETER			B OUTSIDE DIAMETER			C LENGTH		⑪ RAD	⑫ MATING BEARING
	NOM	MIN	MAX	NOM	MIN	MAX	MIN	MAX		
MS 17136-1	3/8	.3745	.3750	9/16	.5620	.5625	.505	.510	.025	MS 17131 -16
MS " -2	3/8	.3745	.3750	9/16	.5620	.5625	.755	.760	.025	MS " -17
MS " -3	3/8	.3745	.3750	9/16	.5620	.5625	.777	.784	.025	MS " -17
MS " -4	3/8	.3745	.3750	5/8	.6245	.6250	.505	.510	.025	MS " -18
MS " -5	3/8	.3745	.3750	5/8	.6245	.6250	.755	.760	.025	MS " -19
MS " -6	7/16	.4370	.4375	5/8	.6245	.6250	.777	.784	.040	MS " -19
MS " -7	1/2	.4995	.5000	3/4	.7495	.7500	.505	.510	.040	MS " -22
MS " -8	1/2	.4995	.5000	3/4	.7495	.7500	.755	.760	.040	MS " -23
MS " -9	1/2	.4995	.5000	3/4	.7495	.7500	.777	.784	.040	MS " -23
MS " -10	5/8	.6245	.6250	7/8	.8745	.8750	.755	.760	.040	MS " -26
MS " -11	5/8	.6245	.6250	7/8	.8745	.8750	1.005	1.010	.040	MS " -27
MS " -12	5/8	.6245	.6250	7/8	.8745	.8750	.777	.784	.040	MS " -27
MS " -13	3/4	.7495	.7500	1	.9995	1.0000	.755	.760	.040	MS " -29
MS " -14	3/4	.7495	.7500	1	.9995	1.0000	1.005	1.010	.040	MS " -30
MS " -15	3/4	.7495	.7500	1	.9995	1.0000	1.027	1.034	.040	MS " -30
MS " -16	13/16	.8120	.8125	1	.9995	1.0000	.755	.760	.040	MS " -29
MS " -17	13/16	.8120	.8125	1	.9995	1.0000	1.005	1.010	.040	MS " -30
MS " -18	7/8	.8745	.8750	1-1/8	1.1245	1.1250	1.005	1.010	.040	MS " -33
MS " -19	7/8	.8745	.8750	1-1/8	1.1245	1.1250	1.027	1.034	.040	MS " -33
MS " -20	15/16	.9370	.9375	1-1/8	1.1245	1.1250	1.005	1.010	.040	MS " -33
MS " -21	1	.9995	1.0000	1-1/4	1.2495	1.2500	1.005	1.010	.040	MS " -35
MS " -22	1	.9995	1.0000	1-1/4	1.2495	1.2500	1.027	1.034	.040	MS " -35
MS " -23	1	.9995	1.0000	1-1/4	1.2495	1.2500	1.255	1.260	.040	MS " -36
MS " -24	1-1/8	1.1245	1.1250	1-3/8	1.3745	1.3750	1.255	1.260	.040	MS " -40
MS " -25	1-3/16	1.1870	1.1875	1-1/2	1.4995	1.5000	1.255	1.260	.040	MS " -43
MS " -26	1-1/4	1.2495	1.2500	1-1/2	1.4995	1.5000	1.005	1.010	.060	MS " -42
MS " -27	1-1/4	1.2495	1.2500	1-1/2	1.4995	1.5000	1.277	1.284	.060	MS " -43
MS " -28	1-1/4	1.2495	1.2500	1-1/2	1.4995	1.5000	1.255	1.260	.060	MS " -43
MS " -29	1-3/8	1.3745	1.3750	1-5/8	1.6245	1.6250	1.255	1.260	.060	MS " -45
MS " -30	1-7/16	1.4370	1.4375	1-3/4	1.7495	1.7500	1.505	1.510	.060	MS " -48
MS " -31	1-1/2	1.4995	1.5000	1-3/4	1.7495	1.7500	1.005	1.010	.060	MS " -47
MS " -32	1-1/2	1.4995	1.5000	1-3/4	1.7495	1.7500	1.505	1.510	.060	MS " -48
MS " -33	1-7/8	1.8745	1.8750	2-1/8	2.1245	2.1250	1.505	1.510	.060	MS " -54
MS " -34	2	1.9995	2.0000	2-1/2	2.4995	2.5000	1.505	1.510	.060	MS " -57
MS 17136-35	2-1/2	2.4995	2.5000	2-3/4	2.7495	2.7500	1.005	1.010	.060	MS 17131 -59

APPROVED 8 FEB 1960

REVISED

CUSTODIANS	OTHER INT.		
P.A.–ORD	A- CE,TC,QMC, SigC,CmlC.	**MILITARY STANDARD**	**MS 17136**
NAVY–NORD	N-	RACE, BEARING, INNER-CARBON STEEL, NEEDLE ROLLER BEARING	
AF–USAF	AF-		
PROCUREMENT SPECIFICATION NONE		SUPERSEDES:	SHEET 1 OF 2

DD FORM 1 OCT 52 672-1

THIS MILITARY STANDARD IS APPROVED BY THE DEPARTMENT OF DEFENSE AND IS MANDATORY ON ALL ACTIVITIES SELECTION FOR ALL NEW ENGINEERING AND DESIGN APPLICATIONS AND FOR REPETITIVE USE SHALL BE MADE FROM THIS DOCUMENT.

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FED. SUP CLASS
3110

MATERIAL:

1. STEEL, CARBON, CARBURIZING GRADE, IN ACCORDANCE WITH 1010 OF FEDERAL STANDARD 66.

HEAT TREATMENT:

2. CASE HARDENED TO ROCKWELL "C" 58 MINIMUM.

SURFACE FINISH:

3. SURFACE MARKED $\overset{16}{\checkmark}$ SHALL HAVE MAXIMUM SURFACE ROUGHNESS AS INDICATED, PER MIL-STD-10.

FINISH (PROTECTIVE):

4. PLAIN (NOT PLATED).
5. COATING OF RUST PREVENTIVE TO BE APPLIED BY MANUFACTURER.

DIMENSIONS:

6. DIMENSIONS ARE IN INCHES.

GENERAL:

7. INTENDED FOR USE ON UNHARDENED SHAFTS IN CONJUNCTION WITH BEARING, ROLLER, NEEDLE: THIN OUTER RACE, WITHOUT INNER RACE, MS 17131.
8. INTENDED FOR USE ON SHAFT WHOSE MAXIMUM DEFLECTION IS .0005 INCHES PER INCH OF BEARING LENGTH.
9. NUMBER OF OIL HOLES FROM A MINIMUM OF ONE (1) TO A MAXIMUM OF FOUR (4) IS OPTIONAL; DIAMETER TO CONFORM WITH MANUFACTURERS STANDARD PRACTICE.
10. OIL GROOVE OPTIONAL; WHERE PROVIDED, GROOVE WIDTH TO CONFORM WITH MANUFACTURERS STANDARD PRACTICE.
11. RACE MUST CLEAR THE MAX. SHAFT FILLET RADIUS SHOWN IN "RAD" COLUMN. THE RADII OR CHAMFERS ON THE INSIDE DIAMETER OF THE INNER RACE MAY BE UNEQUAL. WHERE THE RADII ON THE INSIDE DIAMETER ARE UNEQUAL, THE SURFACE WITH THE LARGER RELIEF SHOULD BE MOUNTED AGAINST THE SHAFT SHOULDER.
12. GIVEN THE SHAFT DIAMETER AND THE RADIAL LOAD FOR A PARTICULAR BEARING APPLICATION, THE BORE DIAMETER OF THE INNER RACE IS SELECTED FROM THE SHAFT SIZE. THE OUTSIDE DIAMETER AND LENGTH ARE CHOSEN BY DIMENSIONALLY MATING THE INNER RACE WITH THE NEEDLE CONTAINING OUTER RACE ON MS 17131 WHICH EXHIBITS THE REQUIRED LOAD CAPACITY.
13. MARKING SHALL CONSIST OF THE MS PART NUMBER AND THE MANUFACTURER'S IDENTIFICATION IN ACCORDANCE WITH MIL-STD-130. ON INNER RACES WHERE THE INSIDE DIAMETER RELIEF RADII ARE NOT EQUAL, THE MARKING SHALL APPEAR ON THE SURFACE WITH THE LESSER RADIUS.
14. THIS STANDARD TAKES PRECEDENCE OVER DOCUMENTS REFERENCED HEREIN.
15. REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATION FOR BID.

APPROVED 8 FEB 1960 REVISED

CUSTODIANS P.A.-ORD NAVY-NORD AF-USAF	OTHER INT. A- CE,TC,QMC, SigC,CmlC N- AF-	**MILITARY STANDARD** RACE, BEARING, INNER, CARBON STEEL, NEEDLE ROLLER BEARING	**MS 17136**
PROCUREMENT SPECIFICATION NONE		SUPERSEDES:	SHEET 2 OF 2

DD FORM 1 OCT 52 672-1

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